

EXPERIMENT 9

Student Name: ARYAN DAHIYA

Branch: AIT-CSE-AIML

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Subject Name: ADBMS

UID: 24BAI70697

Section/Group: 24AIT_KRG1_B

Subject Code: 24CSP-310

Question 1

Question:

An organization stores employee details in the employee_info table. Create a PostgreSQL stored procedure that accepts an employee's **gender** as an input parameter and calculates the total number of employees belonging to that gender. The calculated count should be returned through an **OUT parameter** and displayed using RAISE NOTICE.

Solution:

```
CREATE OR REPLACE PROCEDURE sp_proc(  
    IN p_gender VARCHAR,  
    OUT p_tot_cnt INT  
)  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    SELECT COUNT(*)  
    INTO p_tot_cnt  
    FROM employee_info  
    WHERE gender = p_gender;  
  
    RAISE NOTICE 'Total count of % is %',  
        p_gender, p_tot_cnt;  
END;
```

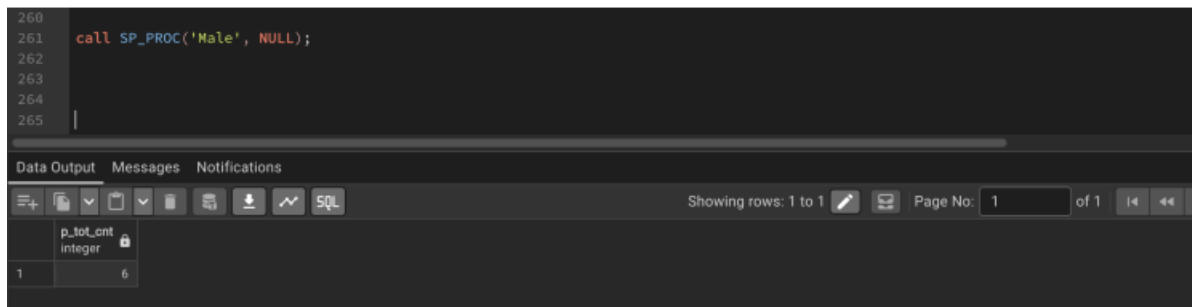
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Calling the procedure:

```
CALL sp_proc('Male', NULL);
```

The procedure counts all employees whose gender is **Male**, returns the count through p_tot_cnt, and displays it using RAISE NOTICE.



```
260
261 call SP_PROC('Male', NULL);
262
263
264
265 |
```

Data Output Messages Notifications

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p_tot_cnt	integer
1	6

Question 2

Question:

An e-commerce company maintains product stock information in the products table and completed orders in the sales table. Create a PostgreSQL stored procedure that accepts a **product name** and **required quantity** as input parameters and processes the purchase only when sufficient stock is available. The procedure should calculate the sale amount as:

Sale Price = Product Price × Quantity Ordered

If sufficient stock is unavailable, the procedure should not modify any data and should display an appropriate notification.

Solution:

```
CREATE OR REPLACE PROCEDURE buy_products(
```

```
    IN p_product_name VARCHAR,
```

```
    IN p_quantity INT
```

```
)
```

```
LANGUAGE plpgsql
```

```
AS $$
```

```
DECLARE
```

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```
v_product_code VARCHAR(20);  
v_price FLOAT;  
v_count INT;  
BEGIN  
  
SELECT product_code, price, quantity_remaining  
INTO v_product_code, v_price, v_count  
FROM products  
WHERE product_name = p_product_name;  
  
IF v_count >= p_quantity THEN  
  
    INSERT INTO sales  
    (order_date, product_code, quantity_ordered, sale_price)  
    VALUES  
    (CURRENT_DATE,  
    v_product_code,  
    p_quantity,  
    v_price * p_quantity);  
  
    UPDATE products  
    SET quantity_remaining = quantity_remaining - p_quantity,  
        quantity_sold = quantity_sold + p_quantity  
    WHERE product_code = v_product_code;  
  
    RAISE NOTICE
```

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'PRODUCT SOLD. Order placed successfully for % unit(s) of %.',

p_quantity, p_product_name;

ELSE

RAISE NOTICE

'INSUFFICIENT QUANTITY. Order cannot be processed for % unit(s) of %.',

p_quantity, p_product_name;

END IF;

END;

\$\$;

Calling the procedure:

CALL buy_products('Samsung Galaxy S23 Ultra', 3);

```
338  
339 call pr_buy_products('Samsung Galaxy S23 Ultra', 3);  
340  
341
```

Data Output Messages Notifications

NOTICE: PRODUCT SOLD..! Order placed successfully for 3 unit(s) of Samsung Galaxy S23 Ultra.
CALL

Query returned successfully in 49 msec.